1) Test-Connection and nslookup

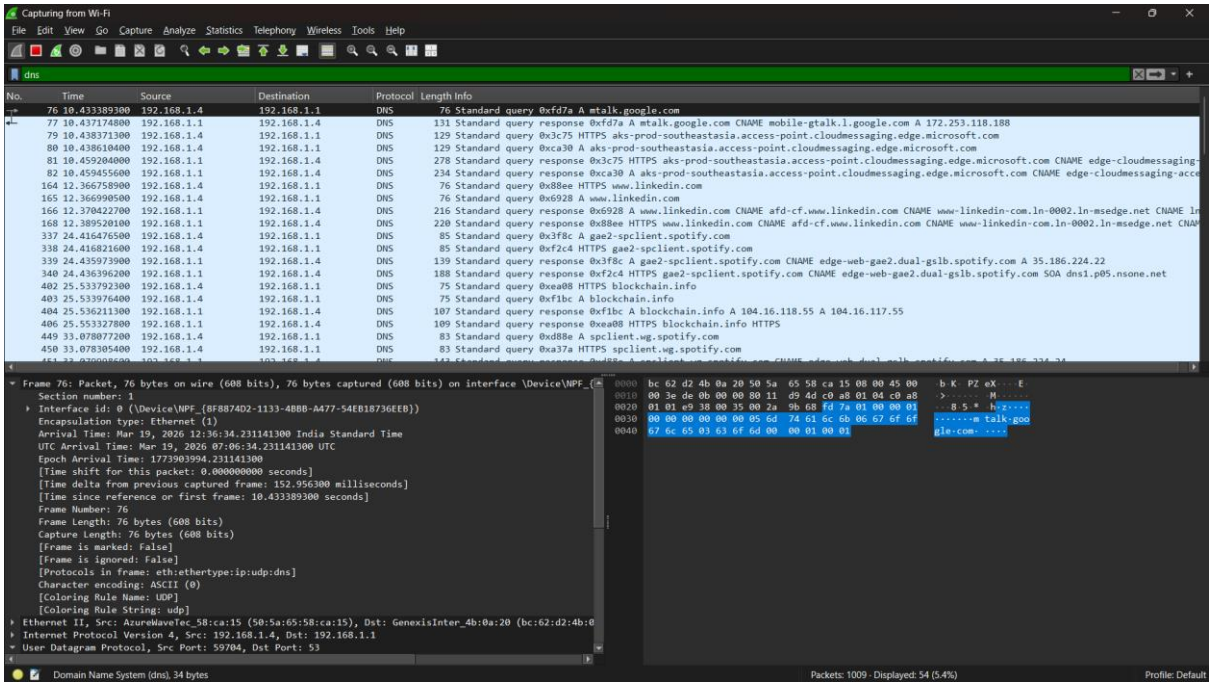
```
PS C:\Users\varssh> Test-Connection www.google.com
>>

Source      Destination    IPV4Address     IPV6Address                    Bytes   Time(ms)
------      -----------    -----------     -----------                    -----   --------
DESKTOPGV   www.google.com 142.251.221.100                                  32      23
DESKTOPGV   www.google.com 142.251.221.100                                  32      43
DESKTOPGV   www.google.com 142.251.221.100                                  32      104
DESKTOPGV   www.google.com 142.251.221.100                                  32      24


PS C:\Users\varssh> Test-Connection www.facebook.com
>>

Source      Destination    IPV4Address     IPV6Address                    Bytes   Time(ms)
------      -----------    -----------     -----------                    -----   --------
DESKTOPGV   www.facebook... 57.144.208.1                                    32      21
DESKTOPGV   www.facebook... 57.144.208.1                                    32      170
DESKTOPGV   www.facebook... 57.144.208.1                                    32      23
DESKTOPGV   www.facebook... 57.144.208.1                                    32      26


PS C:\Users\varssh> Test-Connection www.amazon.com
>>

Source      Destination    IPV4Address     IPV6Address                    Bytes   Time(ms)
------      -----------    -----------     -----------                    -----   --------
DESKTOPGV   www.amazon.com 108.159.19.102                                   32      101
DESKTOPGV   www.amazon.com 108.159.19.102                                   32      25
DESKTOPGV   www.amazon.com 108.159.19.102                                   32      21
```

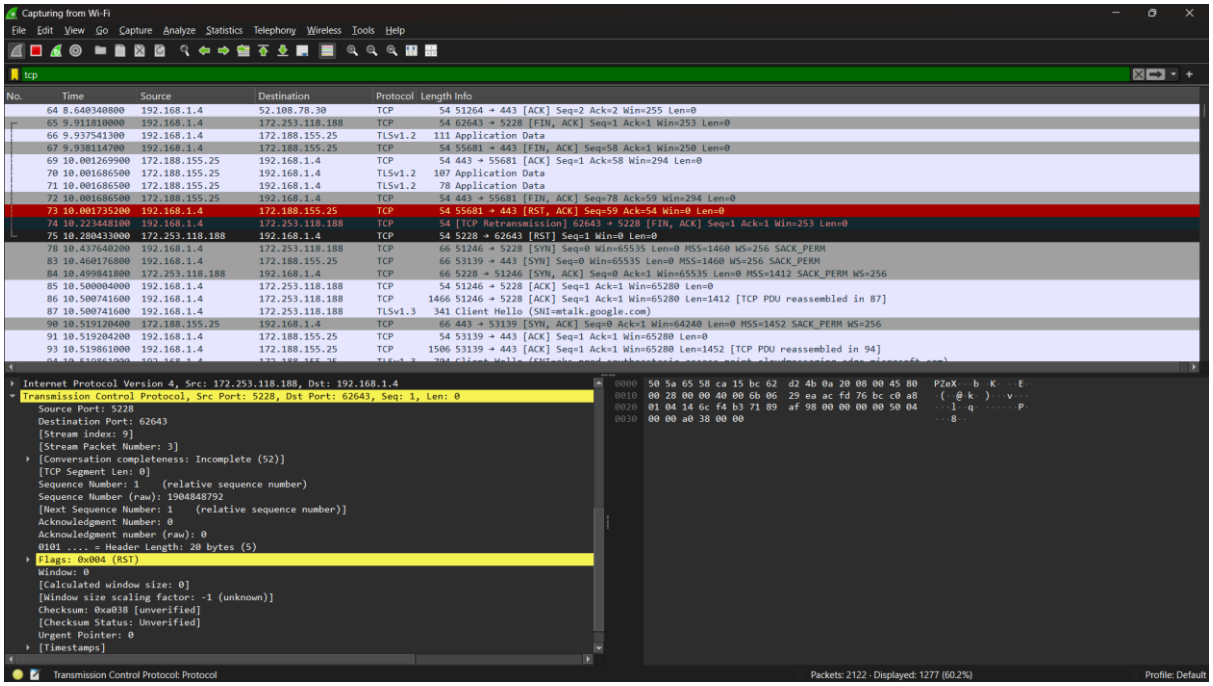
```
DESKTOPGV     www.amazon.com  108.159.19.102                              32      33


PS C:\Users\varssh> Test-Connection www.github.com
>>

Source       Destination     IPV4Address      IPV6Address                      Bytes    Time(ms)
------       -----------     -----------      -----------                      -----    --------
DESKTOPGV     www.github.com  20.207.73.82                                 32      40
DESKTOPGV     www.github.com  20.207.73.82                                 32      56
DESKTOPGV     www.github.com  20.207.73.82                                 32      93
DESKTOPGV     www.github.com  20.207.73.82                                 32      195


PS C:\Users\varssh> Test-Connection www.cisco.com
>>

Source       Destination     IPV4Address      IPV6Address                      Bytes    Time(ms)
------       -----------     -----------      -----------                      -----    --------
DESKTOPGV     www.cisco.com   23.60.108.111                                32      37
DESKTOPGV     www.cisco.com   23.60.108.111                                32      72
DESKTOPGV     www.cisco.com   23.60.108.111                                32      85
DESKTOPGV     www.cisco.com   23.60.108.111                                32      38


PS C:\Users\varssh> nslookup www.cisco.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
```

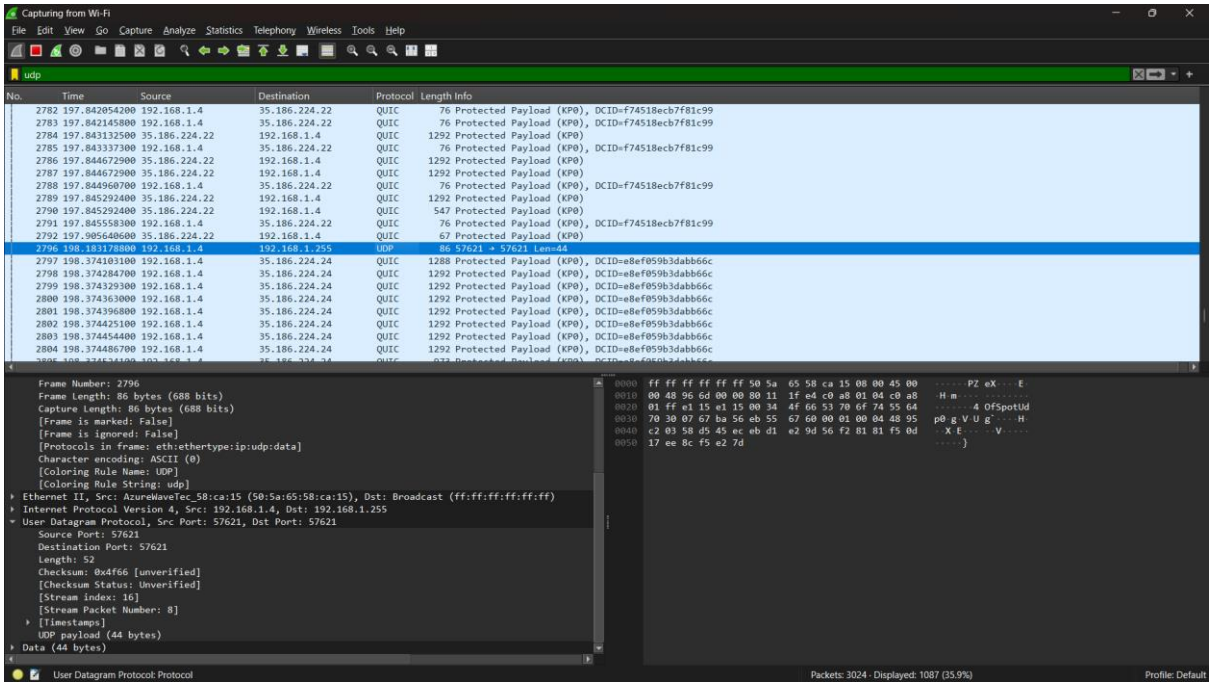
```
Name:    e2867.dsca.akamaiedge.net
Addresses:  2600:1417:53:193::b33
          2600:1417:53:1a9::b33
          23.60.108.111
Aliases:  www.cisco.com
          www.cisco.com.akadns.net
          wwwds.cisco.com.edgekey.net
          wwwds.cisco.com.edgekey.net.globalredir.akadns.net

PS C:\Users\varssh> nslookup www.github.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    github.com
Address:  20.207.73.82
Aliases:  www.github.com

PS C:\Users\varssh> nslookup www.amazon.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    e15316.dsca.akamaiedge.net
Addresses:  2600:1417:53:1ab::3bd4
          2600:1417:53:1bb::3bd4
          23.60.110.84
Aliases:  www.amazon.com
          tp.47cf2c8c9-frontier.amazon.com
```

```
        www.amazon.com.edgekey.net

PS C:\Users\varssh> nslookup www.facebook.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    star-mini.c10r.facebook.com
Addresses:  2a03:2880:f368:1:face:b00c:0:25de
        57.144.208.1
Aliases:  www.facebook.com

PS C:\Users\varssh> nslookup www.google.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    www.google.com
Addresses:  2404:6800:4007:802::2004
        142.251.150.119
        142.251.151.119
        142.251.152.119
        142.251.153.119
        142.251.154.119
        142.251.155.119
        142.251.156.119
        142.251.157.119
```

2)


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
76 10.433389300 192.168.1.4 192.168.1.1 DNS 76 Standard query 0xfd7a A mtalk.google.com
77 10.437174800 192.168.1.1 192.168.1.4 DNS 131 Standard query response 0xfd7a A mtalk.google.com CNAME mobile-gtalk.l.google.com A 172.253.118.188
79 10.438371300 192.168.1.4 192.168.1.1 DNS 129 Standard query 0x3c75 HTTPS aks-prod-southeastasia.access-point.cloudmessaging.edge.microsoft.com
80 10.438610400 192.168.1.1 192.168.1.4 DNS 129 Standard query 0xca30 A aks-prod-southeastasia.access-point.cloudmessaging.edge.microsoft.com
81 10.459204000 192.168.1.1 192.168.1.4 DNS 278 Standard query response 0x3c75 HTTPS aks-prod-southeastasia.access-point.cloudmessaging.edge.microsoft.com CNAME edge-cloudmessaging
82 10.459455600 192.168.1.1 192.168.1.4 DNS 234 Standard query response 0xca30 A aks-prod-southeastasia.access-point.cloudmessaging.edge.microsoft.com CNAME edge-cloudmessaging-acc
164 12.366758900 192.168.1.4 192.168.1.1 DNS 76 Standard query 0x88ee HTTPS www.linkedin.com
165 12.366990500 192.168.1.4 192.168.1.1 DNS 76 Standard query 0x6928 A www.linkedin.com
166 12.370422700 192.168.1.1 192.168.1.4 DNS 216 Standard query response 0x6928 A www.linkedin.com CNAME afd-cf.www.linkedin.com CNAME www-linkedin-com.ln-0002.ln-msedge.net CNAME ln
168 12.389520100 192.168.1.1 192.168.1.4 DNS 220 Standard query response 0x88ee HTTPS www.linkedin.com CNAME afd-cf.www.linkedin.com CNAME www-linkedin-com.ln-0002.ln-msedge.net CNAM
337 24.416476500 192.168.1.4 192.168.1.1 DNS 85 Standard query 0x3f8c A gae2-spclient.spotify.com
338 24.416821600 192.168.1.4 192.168.1.1 DNS 85 Standard query 0xf2c4 HTTPS gae2-spclient.spotify.com
339 24.435973900 192.168.1.1 192.168.1.4 DNS 139 Standard query response 0x3f8c A gae2-spclient.spotify.com CNAME edge-web-gae2.dual-gslb.spotify.com A 35.186.224.22
340 24.436396200 192.168.1.1 192.168.1.4 DNS 188 Standard query response 0xf2c4 HTTPS gae2-spclient.spotify.com CNAME edge-web-gae2.dual-gslb.spotify.com SOA dns1.p05.nsone.net
402 25.533792300 192.168.1.4 192.168.1.1 DNS 75 Standard query 0xea08 HTTPS blockchain.info
403 25.533976400 192.168.1.4 192.168.1.1 DNS 75 Standard query 0xf1bc A blockchain.info
404 25.536211300 192.168.1.1 192.168.1.4 DNS 107 Standard query response 0xf1bc A blockchain.info A 104.16.118.55 A 104.16.117.55
406 25.553327800 192.168.1.1 192.168.1.4 DNS 109 Standard query response 0xea08 HTTPS blockchain.info HTTPS
449 33.078077200 192.168.1.4 192.168.1.1 DNS 83 Standard query 0xd88e A spclient.wg.spotify.com
450 33.078305400 192.168.1.4 192.168.1.1 DNS 83 Standard query 0xa37a HTTPS spclient.wg.spotify.com
Frame 76: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface \Device\NPF_{
Section number: 1
Interface id: 0 (\Device\NPF_{8F8874D2-1133-4BBB-A477-54EB18736EEB})
Encapsulation type: Ethernet (1)
Arrival Time: Mar 19, 2026 12:36:34.231141300 India Standard Time
UTC Arrival Time: Mar 19, 2026 07:06:34.231141300 UTC
Epoch Arrival Time: 1773903994.231141300
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 152.956300 milliseconds]
[Time since reference or first frame: 10.433389300 seconds]
Frame Number: 76
Frame Length: 76 bytes (608 bits)
Capture Length: 76 bytes (608 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:dns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: AzureWaveTec_58:ca:15 (50:5a:65:58:ca:15), Dst: GenexisInter_4b:0a:20 (bc:62:d2:4b:0
Internet Protocol Version 4, Src: 192.168.1.4, Dst: 192.168.1.1
User Datagram Protocol, Src Port: 59704, Dst Port: 53
0000 bc 62 d2 4b 0a 20 50 5a 65 58 ca 15 08 00 45 00
0010 00 3e de 0b 00 00 80 11 d9 4d c0 a8 01 04 c0 a8
0020 01 01 e9 38 00 35 00 2a 9b 68 fd 7a 01 00 00 01
0030 00 00 00 00 00 00 05 6d 74 61 6c 6b 06 67 6f 6f
0040 67 6c 65 03 63 6f 6d 00 00 01 00 01
Domain Name System (dns), 34 bytes
Packets: 1009 · Displayed: 54 (5.4%)
Profile: Default


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
64 8.640340800 192.168.1.4 52.108.78.30 TCP 54 51264 → 443 [ACK] Seq=2 Ack=2 Win=255 Len=0
65 9.911810000 192.168.1.4 172.253.118.188 TCP 54 62643 → 5228 [FIN, ACK] Seq=1 Ack=1 Win=253 Len=0
66 9.937541300 192.168.1.4 172.188.155.25 TLSv1.2 111 Application Data
67 9.938114700 192.168.1.4 172.188.155.25 TCP 54 55681 → 443 [FIN, ACK] Seq=58 Ack=1 Win=250 Len=0
69 10.001269900 172.188.155.25 192.168.1.4 TCP 54 443 → 55681 [ACK] Seq=1 Ack=58 Win=294 Len=0
70 10.001686500 172.188.155.25 192.168.1.4 TLSv1.2 107 Application Data
71 10.001686500 172.188.155.25 192.168.1.4 TLSv1.2 78 Application Data
72 10.001686500 172.188.155.25 192.168.1.4 TCP 54 443 → 55681 [FIN, ACK] Seq=78 Ack=59 Win=294 Len=0
73 10.001735200 192.168.1.4 172.188.155.25 TCP 54 55681 → 443 [RST, ACK] Seq=59 Ack=54 Win=0 Len=0
74 10.223448100 192.168.1.4 172.253.118.188 TCP 54 [TCP Retransmission] 62643 → 5228 [FIN, ACK] Seq=1 Ack=1 Win=253 Len=0
75 10.280433000 172.253.118.188 192.168.1.4 TCP 54 5228 → 62643 [RST] Seq=1 Win=0 Len=0
78 10.437640200 192.168.1.4 172.253.118.188 TCP 66 51246 → 5228 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
83 10.460176800 192.168.1.4 172.188.155.25 TCP 66 53139 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
84 10.499841800 172.253.118.188 192.168.1.4 TCP 66 5228 → 51246 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
85 10.500004000 192.168.1.4 172.253.118.188 TCP 54 51246 → 5228 [ACK] Seq=1 Ack=1 Win=65280 Len=0
86 10.500741600 192.168.1.4 172.253.118.188 TCP 1466 51246 → 5228 [ACK] Seq=1 Ack=1 Win=65280 Len=1412 [TCP PDU reassembled in 87]
87 10.500741600 192.168.1.4 172.253.118.188 TLSv1.3 341 Client Hello (SNI=mtalk.google.com)
90 10.519120400 172.188.155.25 192.168.1.4 TCP 66 443 → 53139 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1452 SACK_PERM WS=256
91 10.519204200 192.168.1.4 172.188.155.25 TCP 54 53139 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
93 10.519861000 192.168.1.4 172.188.155.25 TCP 1506 53139 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1452 [TCP PDU reassembled in 94]
Internet Protocol Version 4, Src: 172.253.118.188, Dst: 192.168.1.4
Transmission Control Protocol, Src Port: 5228, Dst Port: 62643, Seq: 1, Len: 0
Source Port: 5228
Destination Port: 62643
[Stream index: 9]
[Stream Packet Number: 3]
[Conversation completeness: Incomplete (52)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1904848792
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
0101 .... = Header Length: 20 bytes (5)
Flags: 0x004 (RST)
Window: 0
[Calculated window size: 0]
[Window size scaling factor: -1 (unknown)]
Checksum: 0xa038 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
0000 50 5a 65 58 ca 15 bc 62 d2 4b 0a 20 08 00 45 80
0010 00 28 00 00 40 00 6b 06 29 ea ac fd 76 bc c0 a8
0020 01 04 14 6c f4 b3 71 89 af 98 00 00 00 00 50 04
0030 00 00 a0 38 00 00
Transmission Control Protocol: Protocol
Packets: 2122 · Displayed: 1277 (60.2%)
Profile: Default



3) PS C:\Users\varssh\Documents> tracert www.google.com

Tracing route to www.google.com [2001:4860:482b:7700::]

over a maximum of 30 hops:

1 17 ms 4 ms 3 ms 2001:4490:4e05:a42c:b6f9:49ff:feb3:7005

2 * * * Request timed out.

3 7 ms 6 ms 6 ms

PS C:\Users\varssh\Documents> tracert embedur.ai

Tracing route to embedur.ai [162.159.136.54]

over a maximum of 30 hops:

1 3 ms 3 ms 3 ms 192.168.10.1

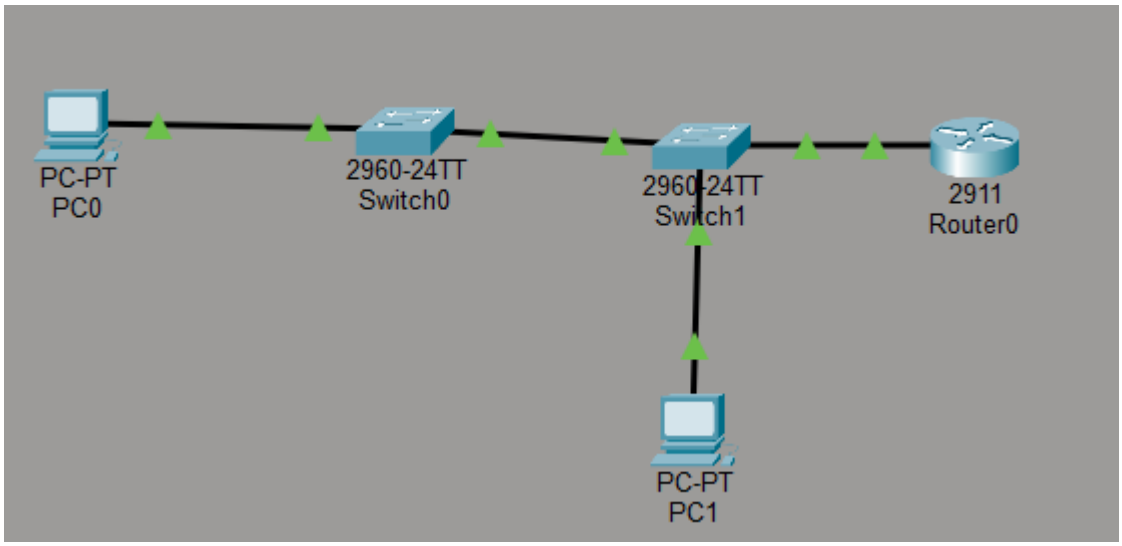
2 6 ms 5 ms 5 ms static.ftth.tcn.117.216.137.1.bsnl.in [117.216.137.1]

3 6 ms 6 ms 6 ms 10.219.5.42

4 * * * Request timed out.

5 * *

5) Trunk Between Switches




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=2ms TTL=128
Reply from 192.168.10.2: bytes=32 time=6ms TTL=128
Reply from 192.168.10.2: bytes=32 time=5ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 6ms, Average = 3ms

C:\>
```

Switch 0 code

Switch>enable

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#interface f0/1

Switch(config-if)#^Z

Switch#

%SYS-5-CONFIG_I: Configured from console by console

Switch#conf t

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#vlan 20
Switch(config-vlan)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface f0/2
Switch(config-if)#switchport mode trunk
```

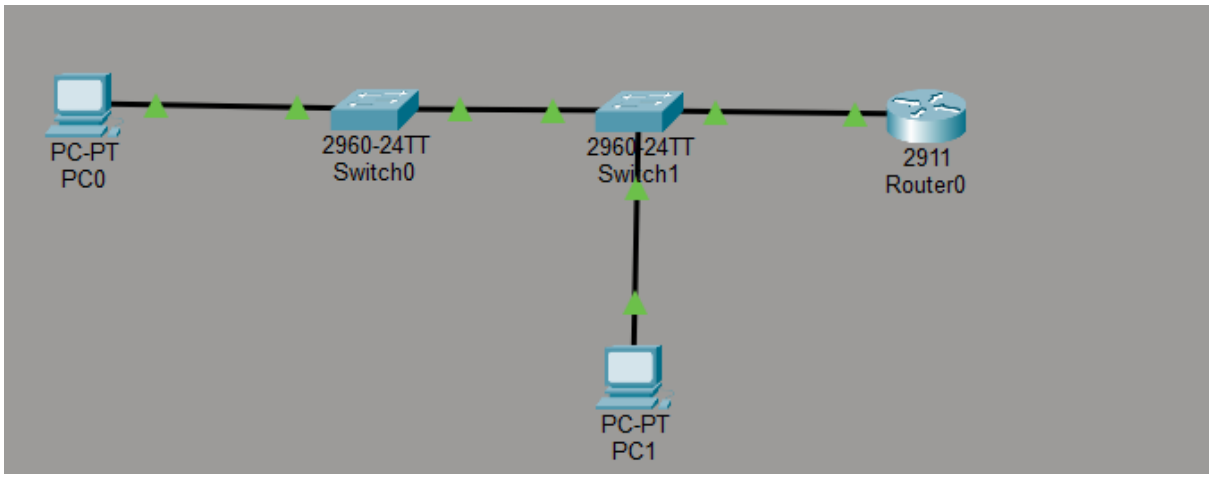
Switch 1 Code

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface f0/1
Switch(config-if)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#vlan 20
Switch(config-vlan)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface f0/2
Switch(config-if)#switchport mode trunk
```

Router 0 Code

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#ip address 192.168.10.1 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g/0.10
                            ^
% Invalid input detected at '^' marker.

Router(config)#interface g0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1
% Incomplete command.
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#interface g0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#end
```

6) Change native vlan



Switch>show interfaces trunk

Port Mode Encapsulation Status Native vlan

Fa0/2 on 802.1q trunking 99

Port Vlans allowed on trunk

Fa0/2 1-1005

Port Vlans allowed and active in management domain

Fa0/2 1,10,20,99

Port Vlans in spanning tree forwarding state and not pruned

Fa0/2 1,10,20,99

VLAN Name Status Ports

---- --------- --------- ---------

1 default active Fa0/4, Fa0/5, Fa0/6, Fa0/7

Fa0/8, Fa0/9, Fa0/10, Fa0/11

Fa0/12, Fa0/13, Fa0/14, Fa0/15

Fa0/16, Fa0/17, Fa0/18, Fa0/19

Fa0/20, Fa0/21, Fa0/22, Fa0/23

Fa0/24, Gig0/1, Gig0/2

10 VLAN0010 active

20 VLAN0020 active Fa0/3

99 VLAN0099 active

1002 fddi-default active

1003 token-ring-default active

1004 fddinet-default active

1005 trnet-default active

Switch#show interfaces trunk

Port Mode Encapsulation Status Native vlan

Fa0/1 on 802.1q trunking 99

Fa0/2 on 802.1q trunking 99

Port Vlans allowed on trunk

Fa0/1 1-1005

Fa0/2 1-1005

Port Vlans allowed and active in management domain

Fa0/1 1,10,20,99

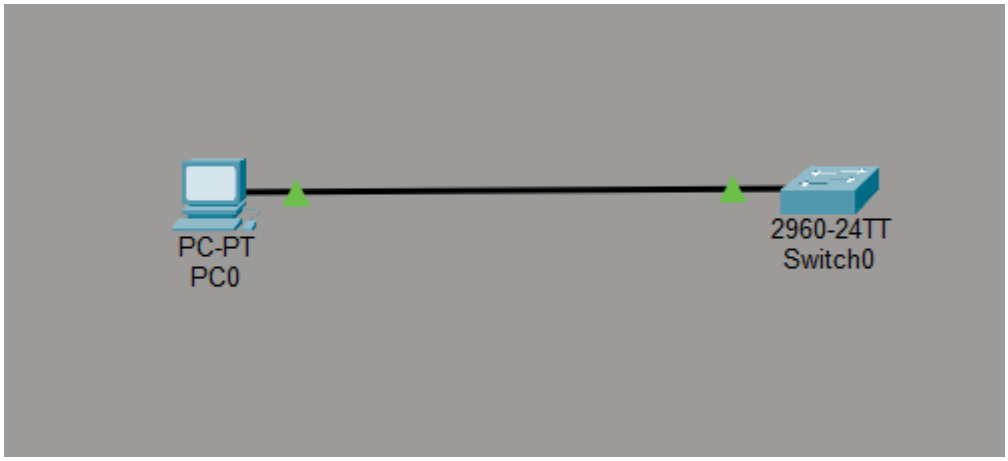
Fa0/2 1,10,20,99

Port Vlans in spanning tree forwarding state and not pruned

Fa0/1 1,10,20,99

Fa0/2 1,10,20,99

7) Telnet to Switch



SWITCH CODE INCLUDING TELNET SETUP

Switch>enable

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#vlan 99

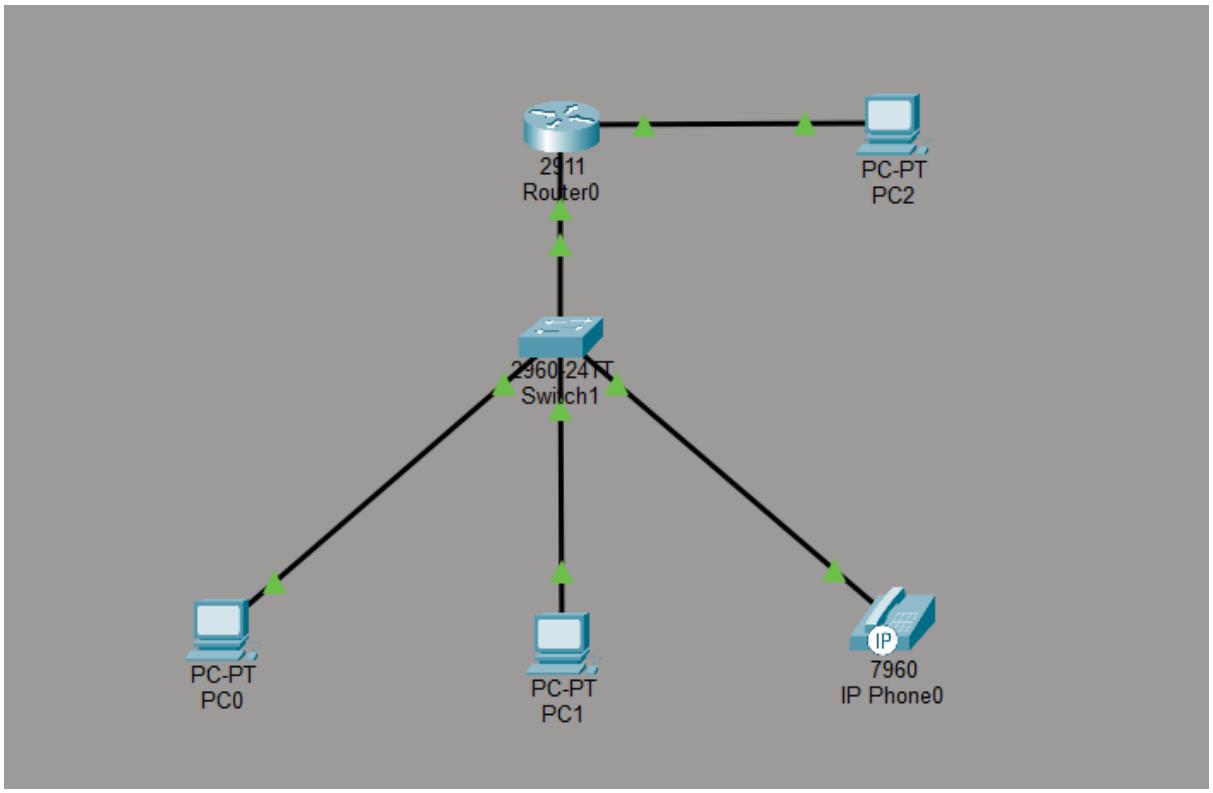
```
Switch(config-vlan)#name MANAGEMENT
Switch(config-vlan)#exit
Switch(config)#interface vlan 99
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up

Switch(config-if)#ip address 192.168.99.2 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#interface f0/1
^
% Invalid input detected at '^' marker.
Switch#interface fa0/1
^
% Invalid input detected at '^' marker.
Switch#enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
no
% Incomplete command.
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#end
```

```
Switch#
%SYS-5-CONFIG_I: Configured from console by console


Switch#ip default-gateway 192.168.99.1
^
% Invalid input detected at '^' marker.
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip default-gateway 192.168.99.1
Switch(config)#line vty 0 4
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#transport input telnet
Switch(config-line)#hostname Switch
Switch(config)#ip domain-name lab.local
Switch(config)#username admin secret cisco123
Switch(config)#crypto key generate rsa
% Please define a hostname other than Switch.
Switch(config)#crypto key generate rsa 1024
^
% Invalid input detected at '^' marker.
Switch(config)#crypto key generate rsa
% Please define a hostname other than Switch.
Switch(config)#crypto key generate rsa 1024
^
% Invalid input detected at '^' marker.
Switch(config)#conf t
^
% Invalid input detected at '^' marker.
Switch(config)#hostname SW1
SW1(config)#ip domain-name lab.local
```

```
SW1(config)#crypto key generate rsa
The name for the keys will be: SW1.lab.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

SW1(config)#line vty 0 4
*Mar 1 0:7:33.810: %SSH-5-ENABLED: SSH 1.99 has been enabled
SW1(config-line)#login local
SW1(config-line)#transport input ssh telnet
^
% Invalid input detected at '^' marker.
SW1(config-line)#exit
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#write memory
Building configuration...
[OK]
SW1#
SW1#
```

```
C:\>ping 192.168.99.2

Pinging 192.168.99.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.99.2: bytes=32 time<1ms TTL=255
Reply from 192.168.99.2: bytes=32 time<1ms TTL=255
Reply from 192.168.99.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.99.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet 192.168.99.2
Trying 192.168.99.2 ...Open


User Access Verification

Username:
% Username:  timeout expired!

[Connection to 192.168.99.2 closed by foreign host]
C:\>telnet 192.168.99.2
Trying 192.168.99.2 ...Open


User Access Verification

Username: admin
Password:
SW1>ssh -l admin 192.168.99.2

Password:

Password:

Password:

[Connection to 192.168.99.2 closed by foreign host]
SW1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual down                  down
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
```

8)Setting up different vlans for pc and VOIP mobile



```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name DATA
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name VOICE
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name MGMT
Switch(config-vlan)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/3
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport voice vlan 20
Switch(config-if)#interface fa0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#interface vlan 99
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up

Switch(config-if)#ip address 192.168.99.2 255.255.255.
                                          ^
% Invalid input detected at '^' marker.

Switch(config-if)#ip address 192.168.99.2 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#ip default-gateway 192.168.99.1
Switch(config)#
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
```

```
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface g0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface g0/0.99
Router(config-subif)#encapsulation dot1Q 99 native
Router(config-subif)#ip address 192.168.99.1 255.255.255.0
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.99, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

Router(config-subif)#interface g0/1
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#interface g0/0
Router(config-if)#ip nat inside
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip nat outside
Router(config-if)#access-list 1 permit 192.168.0.0 0.0.255.255
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#
```

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Request timed out.
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time=1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

```
Switch>show vlan brief

VLAN Name                             Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   DATA                             active    Fa0/1, Fa0/2, Fa0/3
20   VOICE                            active    Fa0/3
99   MGMT                             active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>
```

```
Switch>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/4       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/4       1-1005

Port        Vlans allowed and active in management domain
Fa0/4       1,10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/4       1,10,20,99

Switch>
```

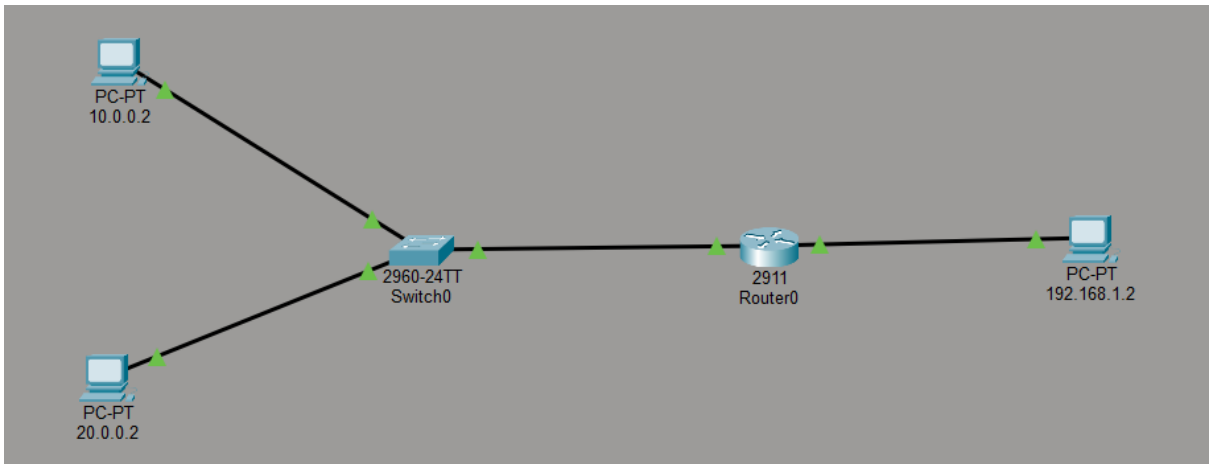
8)

The lack of communication between VLAN 10 and VLAN 20 is expected behavior because VLANs operate as separate Layer 2 broadcast domains, meaning devices in different VLANs cannot exchange traffic directly through a switch. Even if both PCs are correctly assigned to their respective VLANs, the switch will only forward frames within the same VLAN and will not perform any Layer 3 routing between them. To enable communication, an external routing mechanism such as a router configured with inter-VLAN routing (commonly using router-on-a-stick) or a Layer 3 switch with switched virtual interfaces (SVIs) must be implemented so that traffic can be routed between VLAN 10 and VLAN 20.

9)

In a network where VLAN 10 and VLAN 20 are configured on a switch, devices in different VLANs cannot communicate with each other by default because VLANs operate at Layer 2 (Data Link Layer) and isolate broadcast domains. This means that traffic from one VLAN is not forwarded to another VLAN unless a Layer 3 device is introduced. The issue arises when there is no inter-VLAN routing configured, even though VLANs and ports are correctly assigned. To resolve this problem, a router or Layer 3 switch must be used to perform Inter-VLAN Routing, typically through methods like Router-on-a-Stick, where subinterfaces are created for each VLAN and traffic is routed between them. Once routing is enabled and proper gateway addresses are configured on the end devices, communication between VLAN 10 and VLAN 20 becomes possible.

10)INTER VLAN ROUTING



SWITCH CODE

enable

conf t

vlan 10

name VLAN10

vlan 20

name VLAN20

```
exit
interface fa0/1
switchport mode access
switchport access vlan 10
no shutdown
interface fa0/2
switchport mode access
switchport access vlan 20
no shutdown
interface fa0/3
switchport mode trunk
no shutdown
end
write memory
```

Router Code

```
enable
conf t
interface g0/0
no shutdown
interface g0/0.10
encapsulation dot1Q 10
ip address 10.0.0.1 255.255.255.0
ip nat inside
interface g0/0.20
encapsulation dot1Q 20
ip address 20.0.0.1 255.255.255.0
ip nat inside
interface g0/1
ip address 192.168.1.1 255.255.255.0
ip nat outside
```

no shutdown

access-list 1 permit 10.0.0.0 0.0.0.255

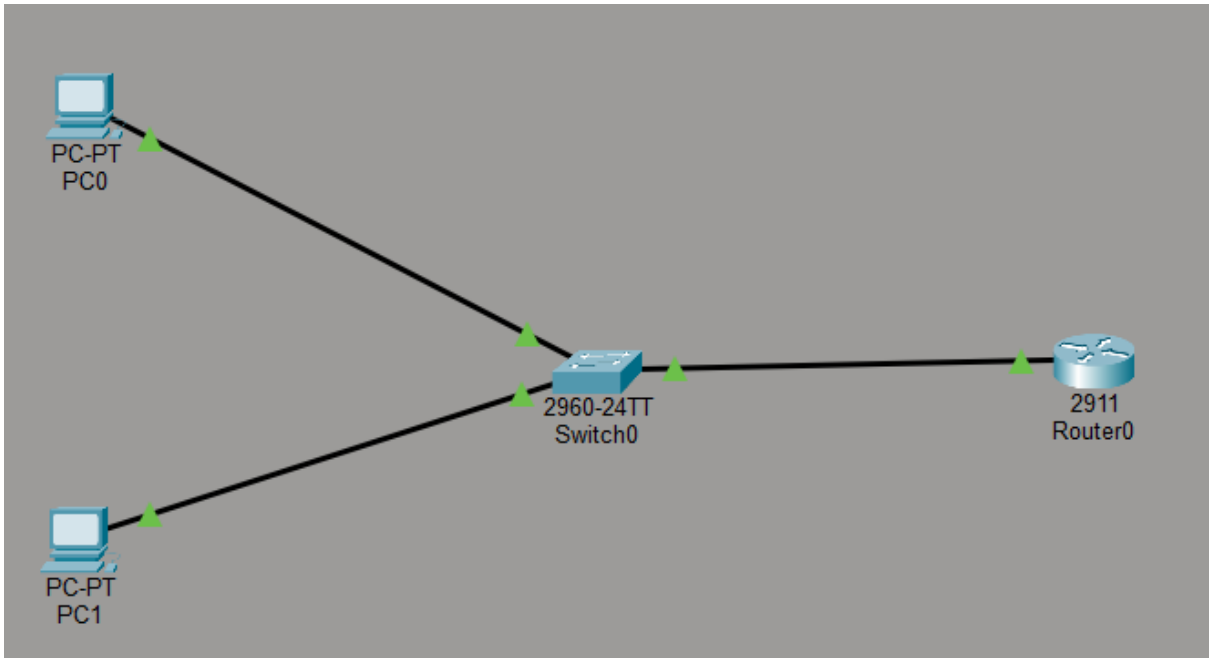
access-list 1 permit 20.0.0.0 0.0.0.255

ip nat inside source list 1 interface g0/1 overload

end

write memory

11) Blocking Traffic Between 2 PCS



Router Code

Router>

Router>enable

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface g0/0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

```
Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.1.1 255
^
% Invalid input detected at '^' marker.
Router(config-subif)#ip address 192.168.1.1 255.255
^
% Invalid input detected at '^' marker.
Router(config-subif)#ip address 192.168.1.1 255.255.255.0
Router(config-subif)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#exit
Router(config)#ip access-list extend GC
Router(config-ext-nacl)#deny ip host 192.168.1.10 host 192.168.2.10
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#interface g0/0.10
Router(config-subif)#ip access-group GC in
```

Router(config-subif)#show access-lists

^

% Invalid input detected at '^' marker.

Router(config-subif)#exit

Router(config)#exit

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#show access-lists

Extended IP access list GC

10 deny ip host 192.168.1.10 host 192.168.2.10 (4 match(es))

20 permit ip any any

Switch Code

```
Switch>
Switch>enable
Switch#con f
% Ambiguous command: "con f"
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#interface f0/3
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
```

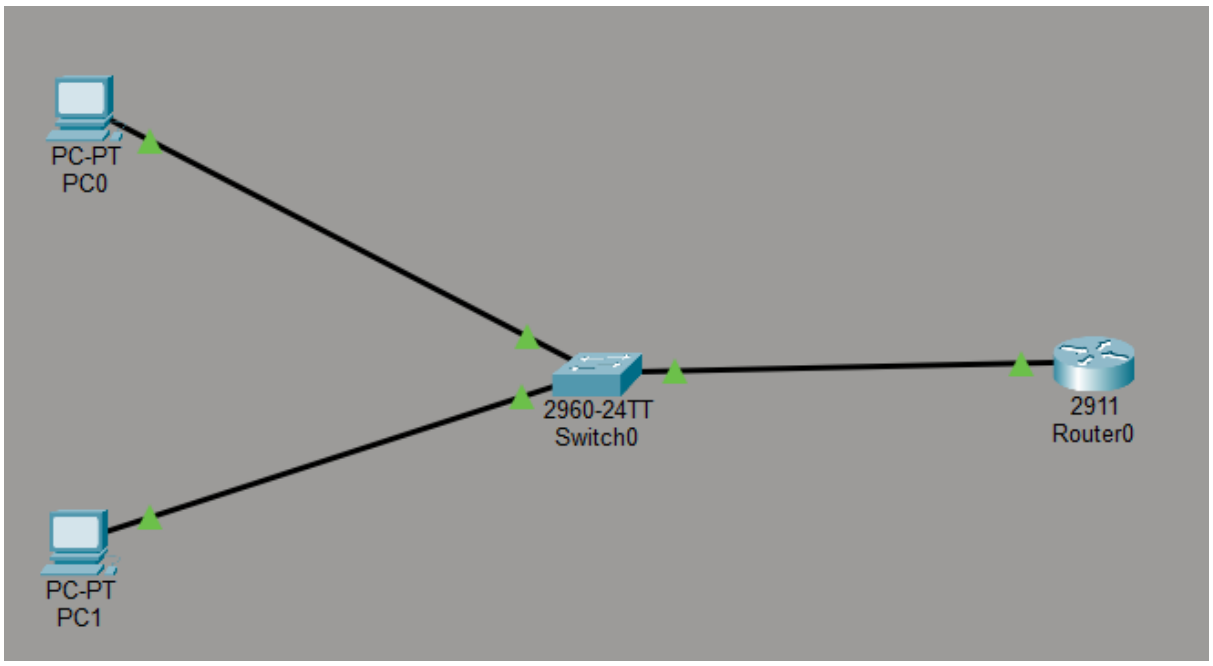
BEFORE CONFIGURING ACL

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

AFTER CONFIGURING ACL

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
C:\>
```

12)



Router CODE added

Router(config)#no ip access-list extended GC

Router(config)#ip access-list extended GV

Router(config-ext-nacl)#deny ip 192.168.1.0 0.0.0.255 host 192.168.2.10

Router(config-ext-nacl)#permit ip any any

Router(config-ext-nacl)#interface g0/0.10

Router(config-subif)#ip access-list GV in

^

% Invalid input detected at '^' marker.

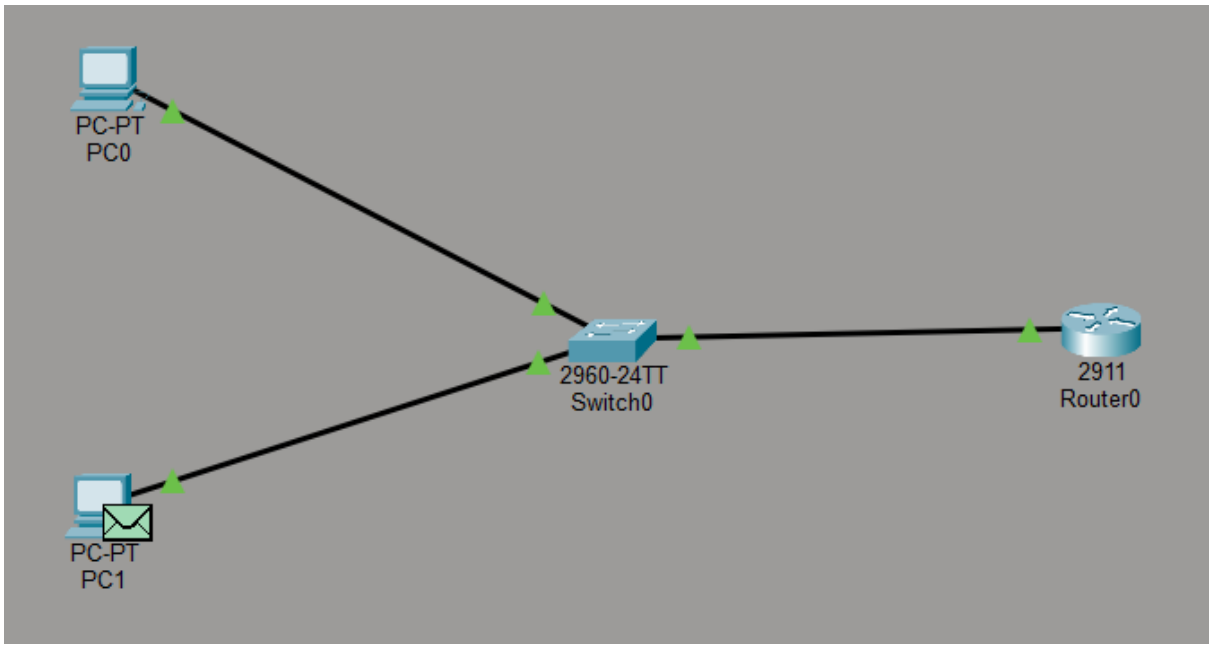
Router(config-subif)#ip access-group GV in

Router(config-subif)#

Router#

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

C:\>
```

13)



Router(config)#access-list 100 deny tcp host 192.168.1.10 host 192.168.2.10 eq 80

Router(config)#access-list 100 permit ip any any

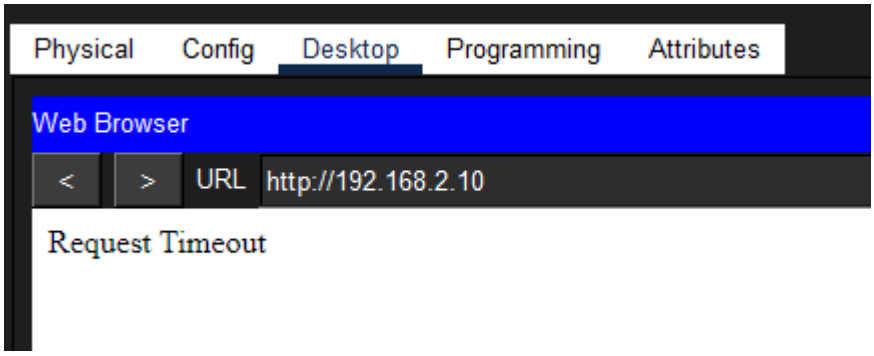
Router(config)#interface g0/0.10

Router(config-subif)#ip access-group 100 in

Router(config-subif)#no ip access-list 100

Router(config-subif)#no access-list 100

Router(config)#



15) **IPERF FROM WSL TO CMD (LOCALHOST)**

WSL OP

vignesh@DESKTOPGV:~$ sudo apt install iperf3 -y

vignesh@DESKTOPGV:~$ iperf3 -s

Server listening on 5201 (test #1)

Accepted connection from 172.23.96.1, port 58070

[5] local 172.23.98.200 port 5201 connected to 172.23.96.1 port 58071

[ID] Interval Transfer Bitrate

[5] 0.00-1.00 sec 743 MBytes 6.23 Gbits/sec

[5] 1.00-2.00 sec 773 MBytes 6.48 Gbits/sec

[5] 2.00-3.00 sec 772 MBytes 6.48 Gbits/sec

[5] 3.00-4.00 sec 747 MBytes 6.26 Gbits/sec

[5] 4.00-5.00 sec 698 MBytes 5.86 Gbits/sec

[5] 5.00-6.00 sec 754 MBytes 6.32 Gbits/sec

[5] 6.00-7.00 sec 744 MBytes 6.24 Gbits/sec

```
[  5]   7.00-8.00   sec   735 MBytes  6.16 Gbits/sec
[  5]   8.00-9.00   sec   748 MBytes  6.27 Gbits/sec
[  5]   9.00-9.17   sec   127 MBytes  6.31 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-9.17   sec  6.68 GBytes  6.26 Gbits/sec                  receiver
-----------------------------------------------------------
Server listening on 5201 (test #2)
```

CMD OP

```
PS C:\Users\guruv\Downloads\iperf3.1.4_64> .\iperf3.exe -c 172.23.98.200
Connecting to host 172.23.98.200, port 5201
[  4] local 172.23.96.1 port 58071 connected to 172.23.98.200 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.00   sec   687 MBytes  5.76 Gbits/sec
[  4]   1.00-2.00   sec   706 MBytes  5.92 Gbits/sec
[  4]   2.00-3.00   sec   702 MBytes  5.89 Gbits/sec
[  4]   3.00-4.00   sec   709 MBytes  5.95 Gbits/sec
[  4]   4.00-5.00   sec   655 MBytes  5.49 Gbits/sec
[  4]   5.00-6.00   sec   652 MBytes  5.47 Gbits/sec
[  4]   6.00-7.00   sec   697 MBytes  5.85 Gbits/sec
[  4]   7.00-8.00   sec   668 MBytes  5.60 Gbits/sec
[  4]   8.00-9.00   sec   672 MBytes  5.64 Gbits/sec
[  4]   9.00-10.00  sec   693 MBytes  5.81 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  6.68 GBytes  5.74 Gbits/sec                  sender
[  4]   0.00-10.00  sec  6.68 GBytes  5.74 Gbits/sec                  receiver
```

iperf Done.